

SOUJANYA PONNAPALLI

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WORK & EDUCATION

University of California, Berkeley

Postdoc · Sky Computing Lab · EECS Department

Supervisors: Prof. Natacha Crooks and Prof. Matei Zaharia

UC Berkeley

2023-Ongoing

University of Texas at Austin

PhD · Systems and Storage Lab · CS Department

Advisor: Prof. Vijay Chidambaram

UT Austin

2017-23

[Minimizing I/O Bottlenecks to Achieve Scalable and High-Throughput Systems](#)

International Institute of Information Technology, Hyderabad

Bachelor's with Honors · SERC Lab · CS and Engineering

Advisor: Prof. Suresh Purini

IIIT-H

2013-17

INTERESTS

Distributed systems · Storage systems · Operating systems · AI systems

I am a systems researcher. I build systems that are not just fast and scalable but also reliable, providing strong guarantees such as performance isolation and fault tolerance in practice.

†Lead postdoc *Project co-leads

PUBLICATIONS

- 1. Metronome: I/O-Efficient Logging for State Machine Replication** [VLDB'27]
Harald Ng*, Soujanya Ponnappalli*†, Kevin Harrison, Reginald Frank, Audrey Cheng, Natacha Crooks, and Paris Carbone
Under revision · [Preprint](#)
- 2. Delta Fair Sharing: Performance Isolation for Multi-Tenant Storage Systems** [VLDB'27]
Tyler Griggs*, Soujanya Ponnappalli*†, Dev Bali, Wenjie Ma, James DeLoye, Audrey Cheng, Jaewan Hong, Natacha Crooks, Scott Shenker, Ion Stoica, and Matei Zaharia
Under submission · [Preprint](#)
- 3. Time is Here for Just-in-Time Systems: Challenges and Opportunities** [MLForSys'26]
Shu Liu, Alexander Krentsel, Shubham Agarwal, Mert Cemri, Ziming Mao, Soujanya Ponnappalli, Alexandros G. Dimakis, Sylvia Ratnasamy, Matei Zaharia, Aditya G. Parameswaran, and Ion Stoica
- 4. Rollbaccine: Herd Immunity against Storage Rollback Attacks in TEEs** [SIGMOD'26]
David C. Y. Chu, Aditya Balasubramanian, Dee Bao, Natacha Crooks, Heidi Howard, Lucky Katanas, and Soujanya Ponnappalli†
- 5. Supporting Our AI Overlords: Redesigning Data Systems to be Agent-First** [CIDR'26]
Shu Liu, Soujanya Ponnappalli, Shreya Shankar, Sepanta Zeighami, Alan Zhu, Shubham Agarwal, Ruiqi Chen, Samion Suwito, Shuo Yuan, Ion Stoica, Matei Zaharia, Alvin Cheung, Natacha Crooks, Joseph E. Gonzalez, and Aditya G. Parameswaran
- 6. SkyStore: Cost-Optimized Object Storage Across Regions and Clouds** [VLDB'25]
Shu Liu, Xiangxi Mo, Moshik Hershcovitch, Henric Zhang, Audrey Cheng, Guy Girmonsky, Gil Vernik, Michael Factor, Tiemo Bang, Soujanya Ponnappalli†, Natacha Crooks, Joseph E. Gonzalez, Danny Harnik, and Ion Stoica
[[Best of VLDB'25 nomination](#)]
- 7. Real Life Is Uncertain. Consensus Should Be Too!** [HotOS'25]
Reginald Frank*, Octavio Lomeli, Neil Giridharan, Soujanya Ponnappalli*†, Marcos K. Aguilera, and Natacha Crooks

PUBLICATIONS

8. **Lost in Translation: The Search for Meaning in Network-Attached AI Accelerator Disaggregation** [HotNets'25]
Jaewan Hong, Yifan Qiao, **Soujanya Ponnappalli**, Shu Liu, Marcos K. Aguilera, Vincent Liu, Christopher J. Rossbach, and Ion Stoica
9. **DINOMO: An Elastic, Scalable, High-Performance Key-Value Store for Disaggregated Persistent Memory** [VLDB'22]
Sekwon Lee, **Soujanya Ponnappalli**, Sharad Singhal, Marcos K. Aguilera, Kimberly Keeton, and Vijay Chidambaram
10. **RainBlock: Faster Transaction Processing in Public Blockchains** [ATC'21]
Soujanya Ponnappalli, Aashaka Shah, Souvik Banerjee, Dahlia Malkhi, Amy Tai, Vijay Chidambaram, and Michael Wei
11. **WineFS: Hugepage-Aware File System for PM that Ages Gracefully** [SOSP'21]
Rohan Kadekodi, Saurabh Kadekodi, **Soujanya Ponnappalli**, Harshad Shirwadkar, Gregory R. Ganger, Aasheesh Kolli, and Vijay Chidambaram
12. **Software-Defined Data Protection: Low Overhead Policy Compliance at the Storage Layer is Within Reach!** [VLDB'21]
Zsolt István, **Soujanya Ponnappalli**, and Vijay Chidambaram
13. **Finding Crash-Consistency Bugs with Bounded Black-Box Crash Testing** [OSDI'18]
Jayashree Mohan, Ashlie Martinez, **Soujanya Ponnappalli**, Pandian Raju, and Vijay Chidambaram
14. **mLSM: Making Authenticated Storage Faster in Ethereum** [HotStorage'18]
Pandian Raju, **Soujanya Ponnappalli**, Evan Kaminsky, Gilad Oved, Zachary Keener, Vijay Chidambaram, and Ittai Abraham
15. **Holographic Storage for the Cloud: Advances and Challenges** [TOS'25]
Nathanaël Cheriére, Jiaqi Chu, Grace Brennan, Pashmina Cameron, Pedro Da Costa, Jannes Gladrow, Guilherme Ilunga, Douglas Kelly, Sarah Lewis, Joowon Lim, Giorgio Maltese, Tony Mason, Greg O'Shea, **Soujanya Ponnappalli**, Michael Rudow, Alan Sanders, Theano Stavrinou, Xingbo Wu, Mengyang Yang, Dushyanth Narayanan, Benn Thomsen, and Antony Rowstron
16. **CrashMonkey and ACE: Systematically Testing File-System Crash Consistency** [TOS'19]
Jayashree Mohan, Ashlie Martinez, **Soujanya Ponnappalli**, Pandian Raju, and Vijay Chidambaram

†Lead postdoc

PREPRINTS

17. **Token Latency Fairness: Performance Isolation for Multi-Tenant LLM Serving** [arXiv'26]
Dev Bali, **Soujanya Ponnappalli**[†], Yichuan Wang, Natacha Crooks, Scott Shenker, and Matei Zaharia
18. **SKYE: Write-Optimized Key-Value Store with Fine-Grained Control over Persistent Memory Accesses** [arXiv'26]
Soujanya Ponnappalli, Sekwon Lee, Rohan Kadekodi, and Vijay Chidambaram

PATENTS

- Recoverable Processes** [US Patent App. 17/981,296]
Jonathan D. Goldstein, Philip A. Bernstein, **Soujanya Ponnappalli**, Jose M. Faleiro, and Peter Charles Shrosbree

ONGOING PROJECTS

Page Cache Fairness: Performance Isolation for Multiple Tenants in Operating Systems

Soujanya Ponnappalli[†], Charisse Ivana Yeung, Natacha Crooks, and Matei Zaharia

Powder: Fine-Grained Fault-Tolerance for Distributed Storage Systems

Reginald Frank*, Soujanya Ponnappalli*[†], Neil Giridharan, Naama Ben-David, and Natacha Crooks

Fugue: Exploring Durability Trade-Offs in Replicated Storage

Diogo Antunes, Baltasar Dinis, Soujanya Ponnappalli, Vijay Chidambaram, Peter Druschel, and Rodrigo Rodrigues

WORK EXPERIENCE

Microsoft Research, Redmond

Mentor: Jonathan Goldstein

Summer'22

Microsoft Research, Redmond

Mentor: Anirudh Badam

Summer'20

Microsoft Research, Cambridge

Mentors: Dushyanth Narayanan and Antony Rowstron

Summer'19

VMware Research, California

Mentors: Michael Wei and Dahlia Malkhi

Summer'18

TEACHING EXPERIENCE

Teaching Assistant at UT Austin

Virtualization with Prof. Vijay Chidambaram

Fall'20,'23

Teaching Assistant at IIIT-H

Algorithms and Data Structures with Prof. Kishore Kothapalli

Operating Systems with Prof. Suresh Purini

Electrical Science with Prof. Rambabu Kalla

2015-17

SERVICE

NSF Proposal Reviewer

2026

Program Committee

OSDI'26, NSDI'26, ATC'25, NSDI'25, EuroSys'25

External Program Committee

FAST'25, ATC'24, NSDI'19

Journal Reviewer, ACM TOCS

2024

Hallway Discussion Lead, SOSF

2021

Shadow PC, EuroSys

2020

Chair, Graduate Application Assistance Program (GAAP@UT)

2020-21

MENTORING

PhD students

Reginald Frank, Shu Liu, Tyler Griggs, Dev Bali, and Diogo Antunes

UC Berkeley

David C. Y. Chu and Jaewan Hong

UC Berkeley

Harald Ng

KTH Sweden

Masters students

James DeLoye, [Extending Delayed Fair Sharing](#)

UC Berkeley

Undergraduate students

Charisse Ivana Yeung

UC Berkeley

Souvik Banerjee, [Implementing Stateless Clients in Ethereum](#)

UT Austin

Gilad Oved, Evan Kaminsky, and Zachary Keener

UT Austin

LEADERSHIP	Organizer, Sky Systems Seminar, Sky Computing Lab, UC Berkeley	2024-25
	Co-organizer, Databases Seminar, Sky Computing Lab, UC Berkeley	2024-25
	Representative, Graduate Association of Computer Sciences (GRACS@UT)	2020-21
	Mentor, Women in Computer Science (WiCS), UT Austin	2019-20
	Co-organizer, Systems Seminar, Lab for Advanced Systems Research, UT Austin	2018
AWARDS & GRANTS	Summer Research Institute (SuRI) Fellowship, EPFL	2025
	Berkeley RDI Frontier Research Funding	2024
	The James C. Browne Graduate Fellowship	2017-18
	IIIT-H Best All-Rounder Gold Medalist	2017
	Dean's Award for ranking in the top 5% of students, IIIT-H	2014-17
REFERENCES	Natacha Crooks and Matei Zaharia (UC Berkeley), Vijay Chidambaram (UT Austin), Marcos K. Aguilera (NVIDIA), and Tianyin Xu (UIUC)	